INDEPENDENT RESEARCH WHITE PAPER

# From Centralized Banking to a Federated Financial Trust Network

A Blockchain-Based Architecture for Digital Identity, KYC/CDD/EDD, Merchant Onboarding and Interoperable Payments in Pakistan

IDENTITY → TRUST → BANKING → MERCHANT → PAYMENT


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Conceptual architecture — not an existing SBP system

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# ES Executive Summary

Pakistan does not need to decentralize banks themselves. It can decentralize the *trust layer* connecting banks, customers, merchants, EMIs, PSPs, PSOs and regulators.

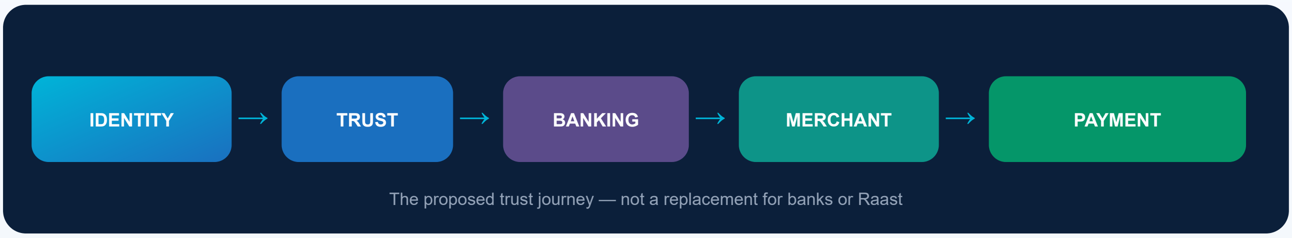


**Figure 0.** The proposed trust journey from identity to payment. PFFTN would sit as a conceptual trust layer — not a replacement for banks or Raast.

1

### What is wrong today?

Customers and merchants often repeat KYC and onboarding across banks, EMIs, PSPs and gateways. Verification is duplicated, data can become inconsistent, and compliance costs rise while inclusion suffers friction.

2

### What has Pakistan already built?

SBP has advanced Shared e-KYC on DLT principles, Raast as an instant payment system, and Raast P2M for merchant acceptance (QR, Alias, IBAN, Request to Pay), plus digital banking and EMI/PSP ecosystems.[1][2][3]

3

### What is being proposed?

4

### Why blockchain / DLT?

**PFFTN** — a conceptual permissioned trust network that stores proofs, attestations, hashes, permissions and state transitions on a ledger, while sensitive data stays off-chain.

Not to become a national database of CNICs. To become a shared machine for proving that something has already been verified — with auditability, consent records and revocation.

### Customers

Could prove identity once via verifiable credentials; authorized institutions verify proofs with consent — while each institution still decides whether to open a relationship.

### Merchants

A proposed **Merchant Financial Identity (MFI)** could support "verify once, reuse securely" across acquiring relationships, aligned with Raast P2M enablement.

### Raast integration

Raast remains the payment rail. PFFTN would be the surrounding identity and trust layer — an extension of existing infrastructure, not a replacement.

### Why it matters

Lower duplicated friction, stronger provenance for AML/CFT workflows, better merchant acceptance economics, and a federated model suited to regulated banking — not permissionless DeFi.

CORE THESIS

"Pakistan does not need to decentralize banks themselves. It can decentralize the TRUST LAYER connecting banks, customers, merchants, EMIs, PSPs, PSOs and regulators."

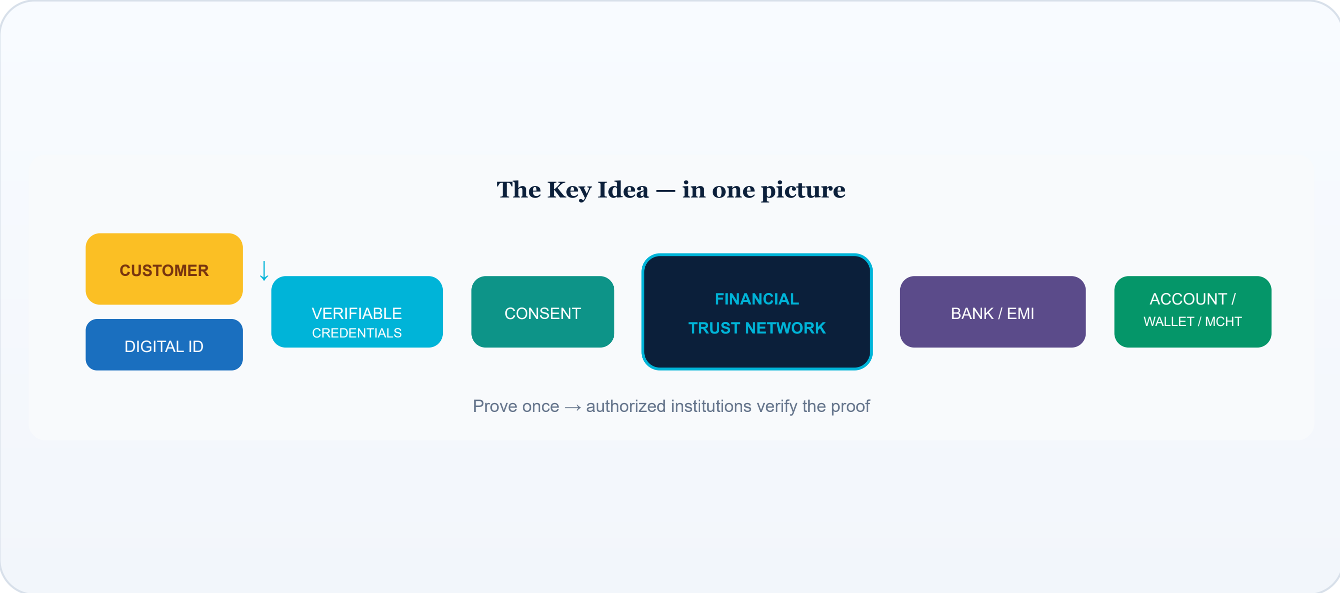


**Figure 1.** Key concept: customer → digital identity → verifiable credentials → consent → trust network → institution → account/wallet/merchant service.

## Contents

## 01 The Problem

CURRENT STATE

Pakistan's digital finance ecosystem has grown quickly — banks, microfinance banks, electronic money institutions (EMIs), payment system operators (PSOs) and payment service providers (PSPs). Growth is good. Fragmented trust is expensive.

**In simple terms:** every time you open a new financial relationship, you often restart the "prove who you are" process — even when another regulated institution already knows you.

### Customer friction

- KYC → Bank A
- KYC → Bank B
- KYC → EMI
- KYC → PSP

Same person. Repeated forms. Repeated waiting.

### Merchant friction

- Onboarding → Bank
- Onboarding → EMI (e.g. wallet acquirer)
- Onboarding → PSP / gateway

Same shop. Many gatekeepers. Slow acceptance.

Duplicated verification creates inconsistent data, higher compliance cost, slower inclusion, and difficulty keeping information current when a customer's life circumstances change.

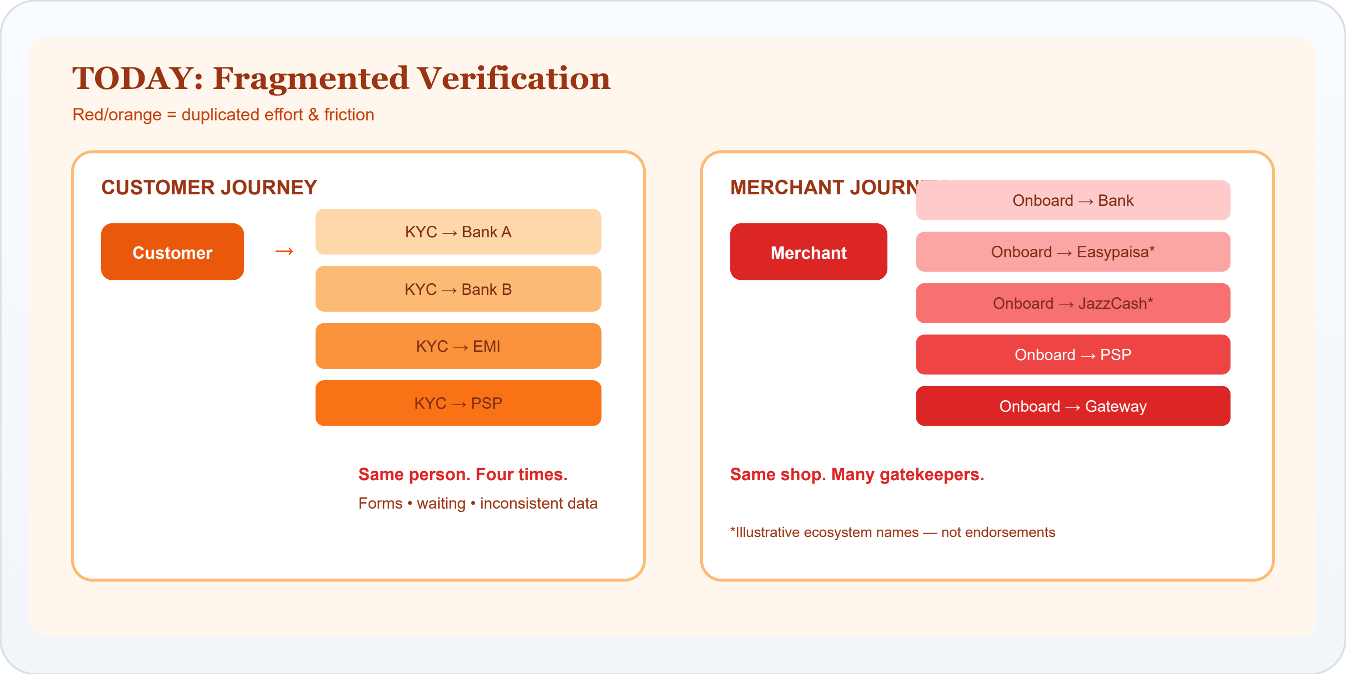


**Figure 2.** The banking problem today — duplicated KYC and merchant onboarding (illustrative).

# 02 The Existing Pakistan Foundation

CURRENT STATE

This proposal does **not** treat Pakistan as starting from zero. Several national building blocks already exist.

### 1. Shared e-KYC

SBP's BPRD Circular Letter No. 22 of 2023 concerns implementation of a Shared Electronic Know Your Customer (e-KYC) Platform.[1] Public descriptions of the initiative state that the platform is built on Distributed Ledger Technology (DLT), enables secure exchange and updating of KYC/CDD information, supports standardization and customer-consent-based access, while underlying KYC/CDD information remains with banks rather than a single central data warehouse.[4][5]

### 2–3. Raast & Raast P2M

Raast is Pakistan's instant payment system. SBP's PSP&OD Circular No. 04 of 2023 launched Raast Person-to-Merchant (P2M) service for regulated entities (banks, MFBs, EMIs, PSOs & PSPs).[2] P2M supports payment acceptance via QR, Raast Alias, IBAN and Request to Pay (RTP).[2][6]

### 4. Digital merchant onboarding

SBP's subsequent PSP&OD Circular Letter No. 01 of 2024 requires that new-to-bank merchants/businesses be enabled for Raast P2M as part of onboarding (directly or via partnered EMI/PSO/PSP), and sets timelines for enabling existing shopping merchants.[3][6]

### 5–7. Ecosystem & consent & DLT

Banks, MFBs, EMIs, PSOs and PSPs already operate under SBP oversight. Shared e-KYC emphasises customer consent. DLT is already part of the national conversation for KYC exchange — which is exactly why a federated trust layer is a natural *extension*, not a moonshot.

### Plain-language takeaway

Pakistan has already built several pieces of the puzzle. What's conceptually missing is a broader **federated trust layer** that can connect identity, compliance attestations, merchant identity and payment readiness — without turning the ledger into a personal-data dump.

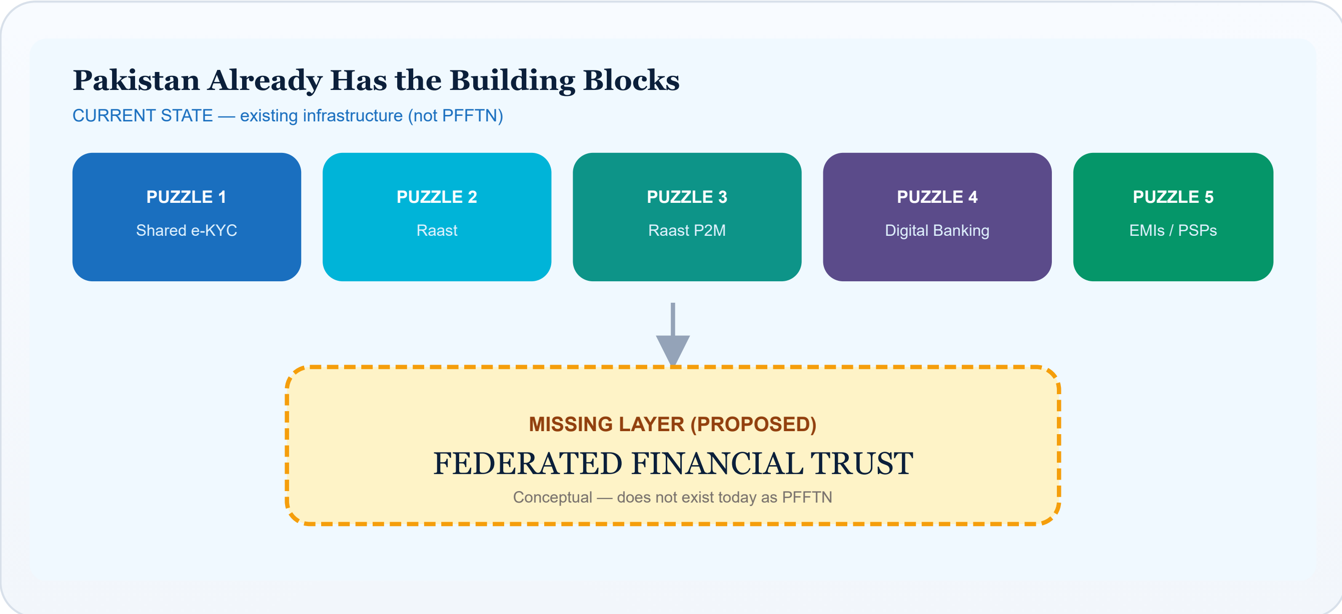


**Figure 3.** Puzzle pieces already in place; federated trust is proposed as the missing layer.

# 03 The New Architecture — PFFTN

PROPOSED ARCHITECTURE

This paper names the conceptual architecture the **Pakistan Federated Financial Trust Network (PFFTN)**. It is *not* an existing SBP system.

**Objective:** create a common trust layer between banks, microfinance banks, EMIs, PSPs, PSOs, merchants, customers, identity providers, regulators and government verification services.

> **In simple terms:** Traditional banking says “every bank checks you separately.” Federated banking says “you prove who you are once, and authorized institutions can verify the proof.” Blockchain/DLT becomes a shared machine for proving that something has already been verified.

Design rule: **store sensitive data off-chain**. Store proofs, attestations, hashes, permissions and state transitions on the ledger.

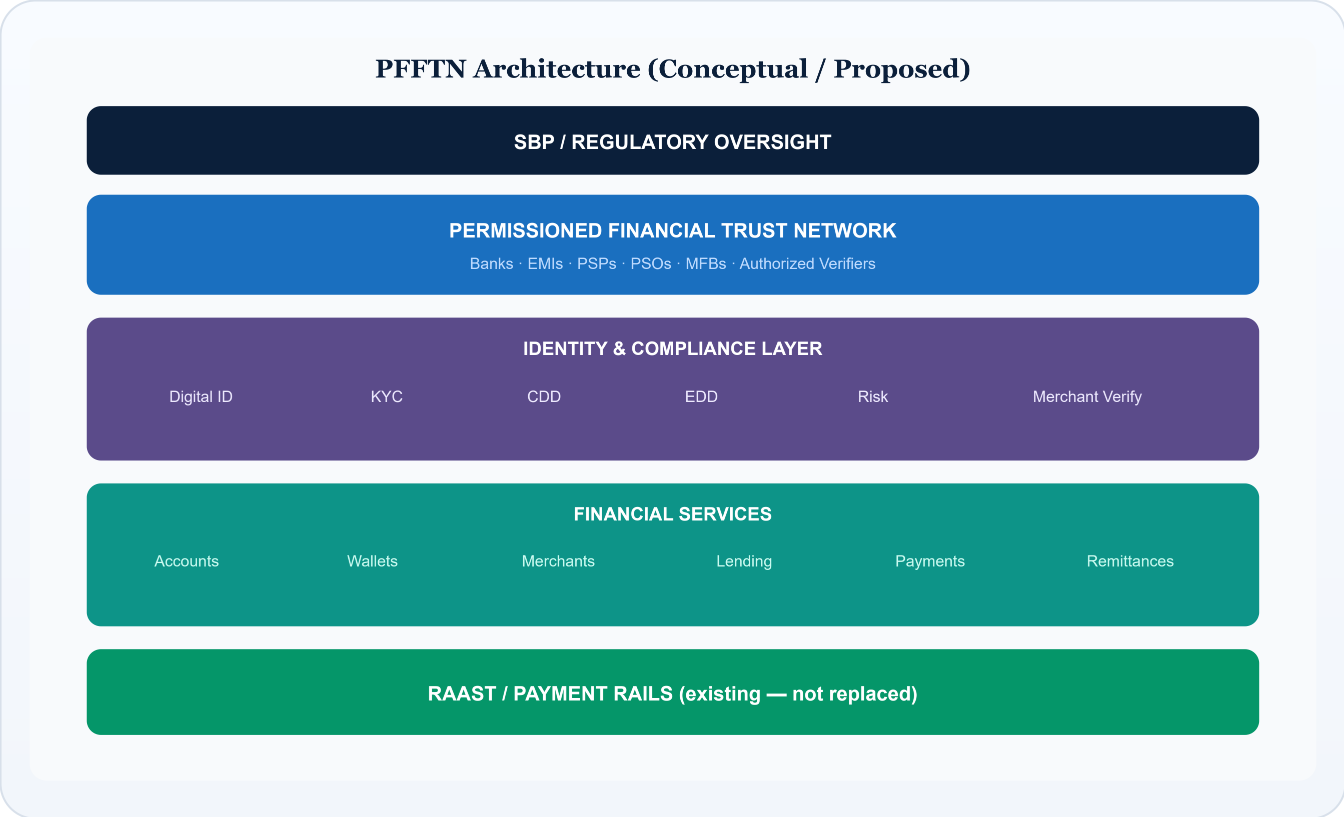


**Figure 4.** Proposed PFFTN layered architecture — regulatory oversight above; Raast rails below.

# 04 Account Opening

PROPOSED ARCHITECTURE

The blockchain would **not** “approve” customers autonomously. Regulated financial institutions remain responsible for onboarding decisions. The network would provide trusted, verifiable evidence that institutions can rely on — subject to law, policy and their own risk appetite.


**Figure 5.** Account opening before (current) and after (proposed credential-assisted flow).

### Responsibility stays with the institution

Even with shared credentials, a bank or EMI must still satisfy its AML/CFT and customer due diligence obligations. PFFTN would be evidence infrastructure — not a liability shield.

# 05 CDD as Distributed Validation

PROPOSED ARCHITECTURE

Customer Due Diligence (CDD) can be viewed as a **network of attestations**, not a single form.

1. Identity Provider → identity verified
2. Bank → customer relationship verified
3. Government / authorized source → relevant attribute verified
4. Compliance system → sanctions screening
5. Risk engine → risk assessment

The ledger would record: **who** verified **what**, **when**, under **which authority**, and whether the verification is **still valid**.

> **In simple terms:** CDD becomes a team sport with signed scorecards — not one bank reinventing the entire checklist every time.

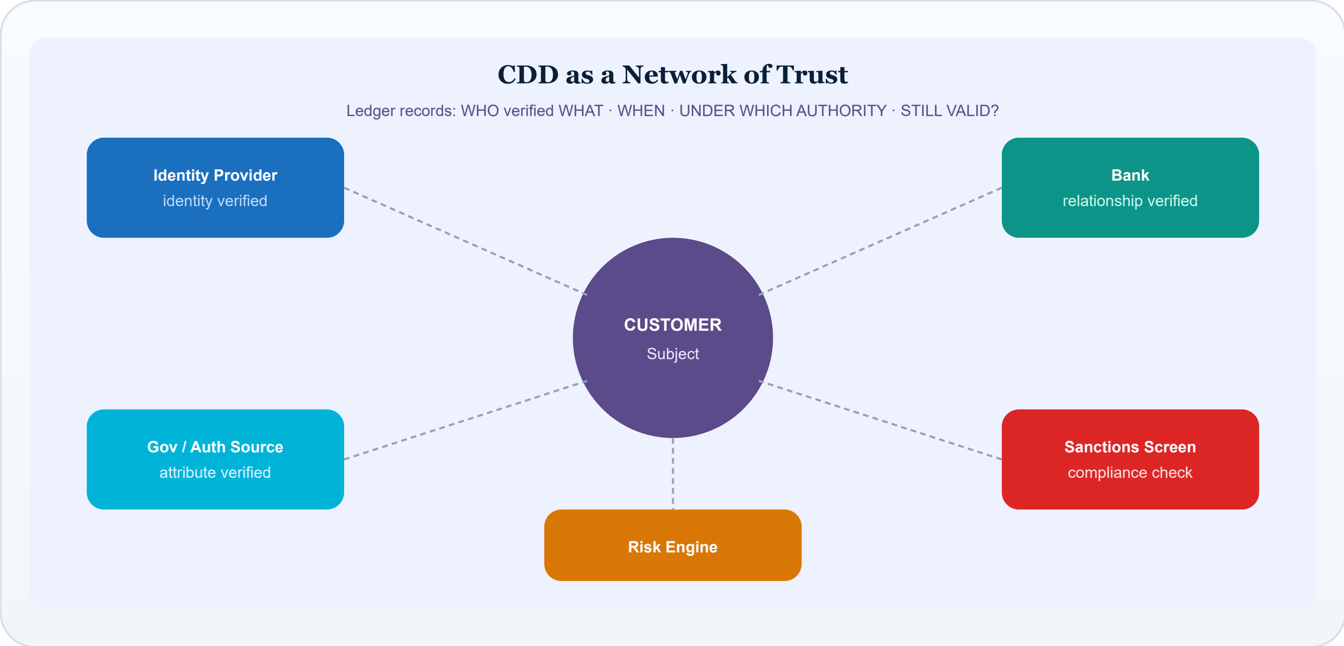


**Figure 6.** CDD as a network of trust — multiple validators, one auditable subject.

# 06 EDD as Enhanced Verification

**PROPOSED ARCHITECTURE**

Enhanced Due Diligence (EDD) should **not** simply mean “more blocks on a chain.”

**EDD = additional evidence + additional verification + stronger controls.**

### Normal / lower-risk path

Identity → KYC → CDD → proportionate controls → ongoing monitoring

### Higher-risk path

Identity → KYC → CDD → risk trigger → EDD → more evidence → enhanced approval → continuous monitoring

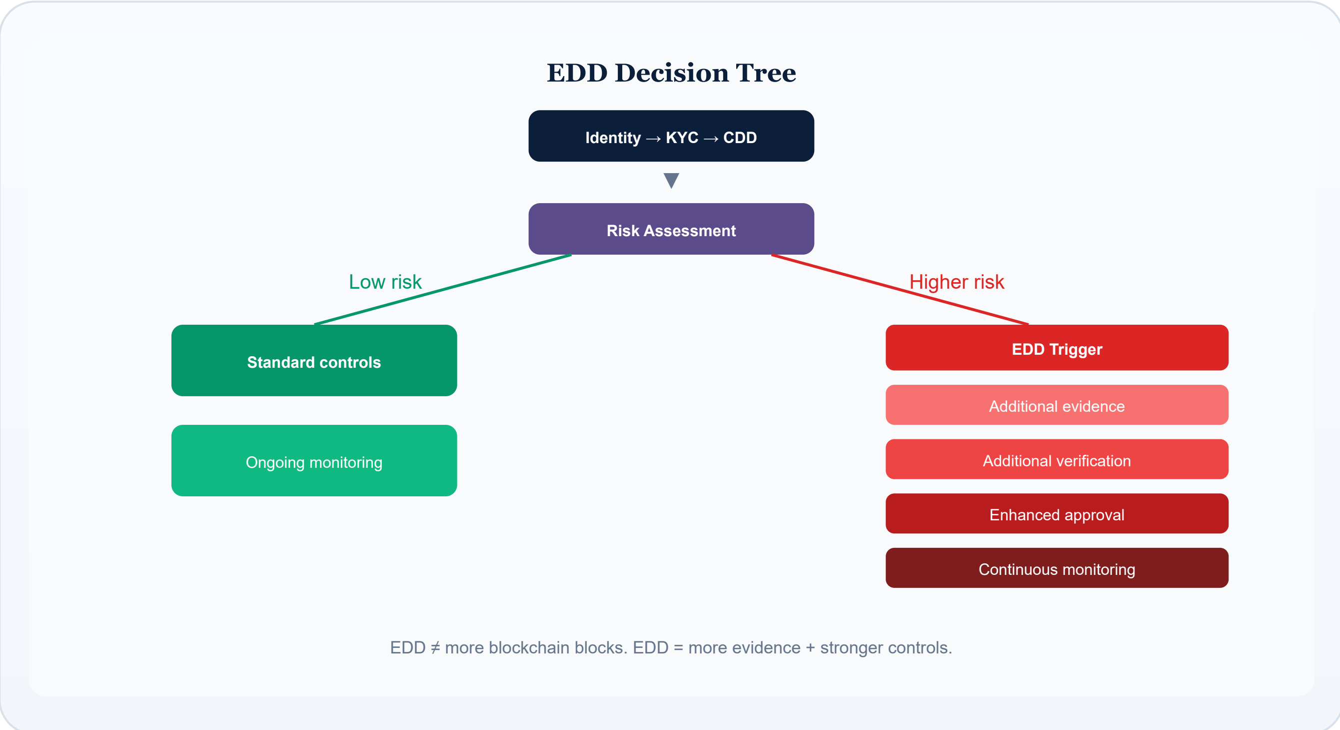


**Figure 7.** EDD decision tree — more evidence and controls, not merely more ledger entries.

# 07 Digital Financial Identity

PROPOSED ARCHITECTURE

A customer could hold a **Financial Identity Credential** — a pack of verifiable attributes such as: identity verified, CNIC verified, address verified, biometric verified, source of income verified, tax status, business/beneficial ownership (where relevant), risk classification, and KYC/CDD/EDD status.

> **Verifiable credential** — in simple terms: a digital proof that says someone has already verified a particular fact. You show the proof; you don't always need to resubmit the original paperwork.

Aligned with privacy-preserving identity patterns discussed in international digital-identity and verifiable-credential work (e.g. W3C Verifiable Credentials data model concepts).[7]

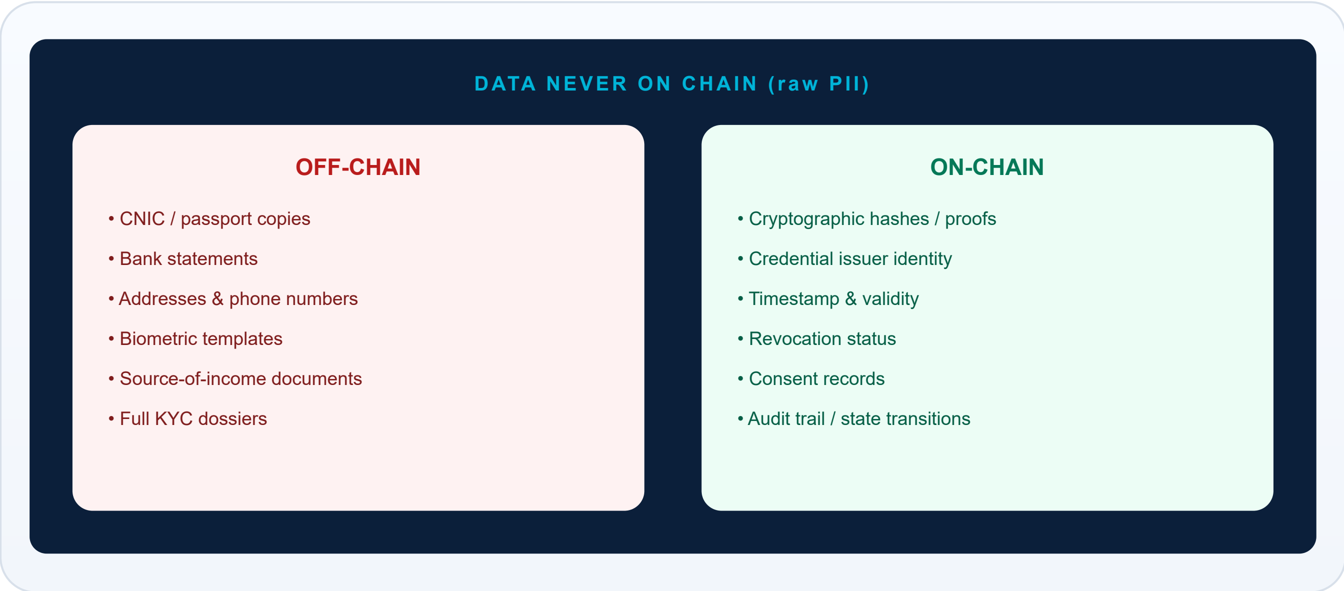


**Figure 8.** On-chain vs off-chain split — raw PII never belongs on the ledger.

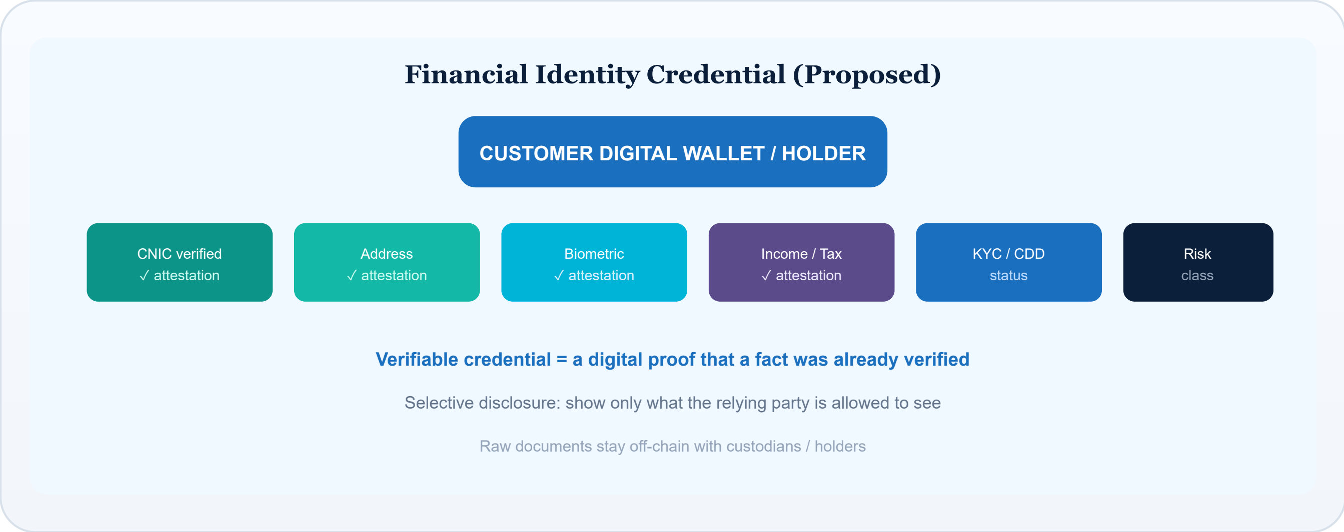


**Figure 9.** Financial Identity Credential — attribute attestations, selective disclosure.

# 08 National Merchant Identity

PROPOSED ARCHITECTURE

One of the highest-leverage ideas in this paper: a reusable **Merchant Financial Identity (MFI)**.

An MFI could conceptually include attestations about legal identity, NTN (where applicable), business registration, beneficial ownership, business category, merchant risk profile, KYC/CDD/EDD status, settlement accounts, Raast merchant identity, QR identity, acquiring relationships and verification history.

**This does not mean every institution receives every piece of data.** Access would be permissioned and consent-driven, subject to law and regulatory requirements.

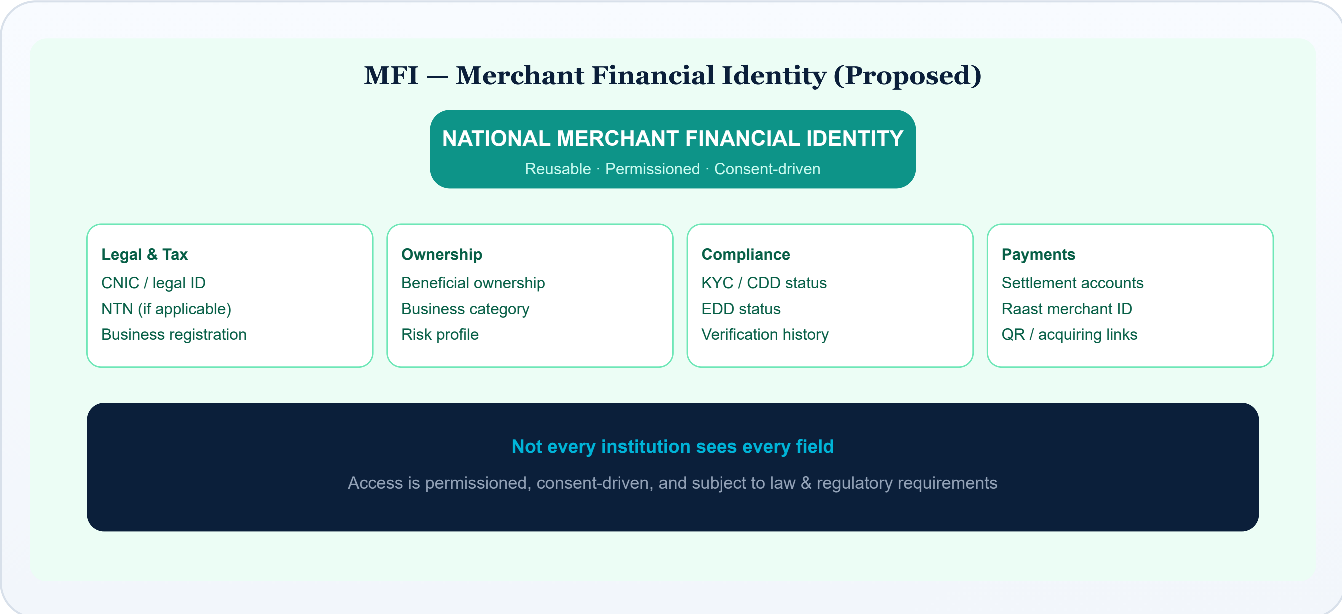


**Figure 10.** Merchant Financial Identity (MFI) — reusable, permissioned merchant trust profile.

# 09 EMI / Bank / PSP Journeys

PROPOSED ARCHITECTURE ILLUSTRATIVE

For illustration, consider a merchant onboarding journey involving an EMI such as Easypaisa. *This does not imply that Easypaisa, JazzCash, any bank, or SBP has agreed to this proposal.*

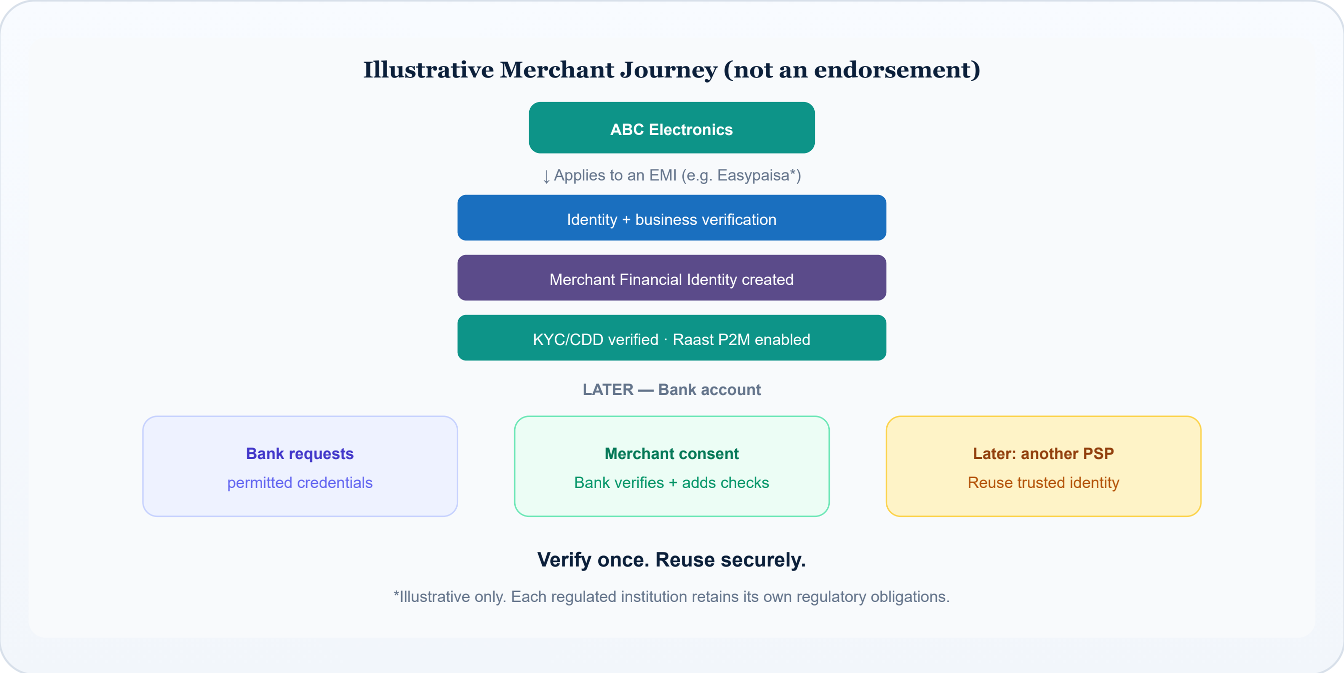


**Figure 11.** Illustrative merchant journey — verify once, reuse securely across relationships.

### Two truths at once

**Verify once. Reuse securely.** And: each regulated institution remains responsible for its own regulatory obligations and final onboarding decision.

# 10 Raast Integration

CURRENT STATE + PROPOSED

**PFFTN is not a replacement for Raast.** Raast remains the payment rail. PFFTN would become the trust and identity layer around the payment ecosystem.

SBP launched Raast P2M to facilitate digital payment acceptance using QR Codes, Raast Alias, IBAN and Request to Pay.[2] Subsequent instructions push broader merchant enablement during onboarding.[3]

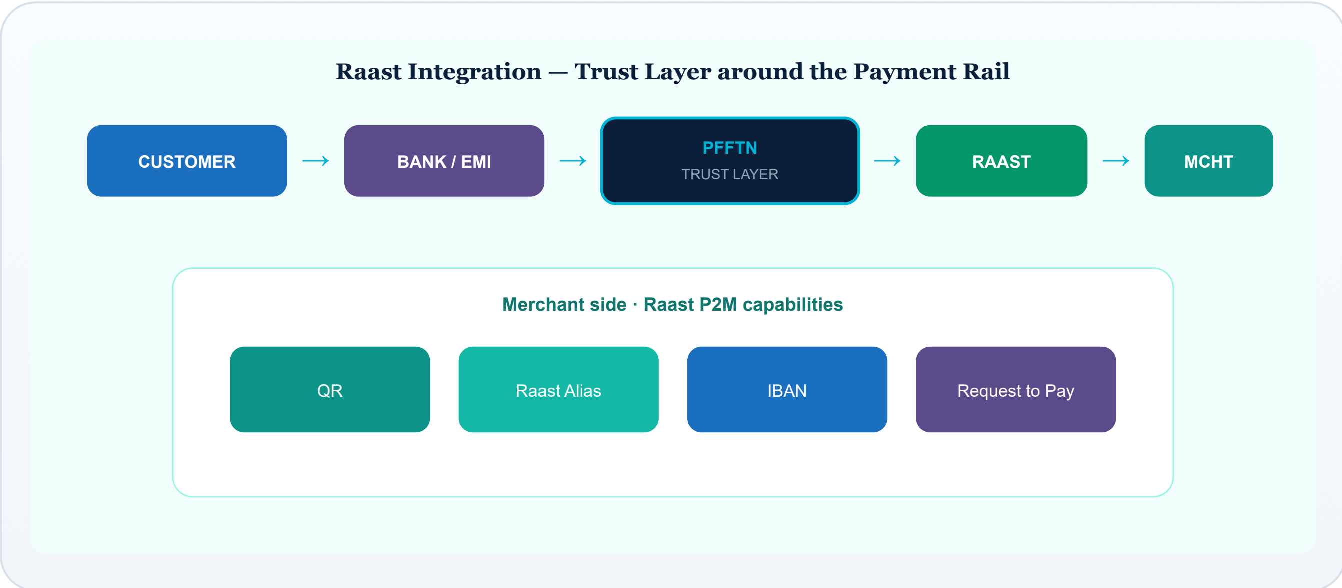


**Figure 12.** Raast stays the rail; PFFTN would wrap identity/trust around payments.

# 11 Merchant Portability

PROPOSED ARCHITECTURE

**Merchant Portability** means a merchant's verified financial identity could travel — with consent and legal authority — across banks, EMIs and PSPs, while each institution retains its own controls.

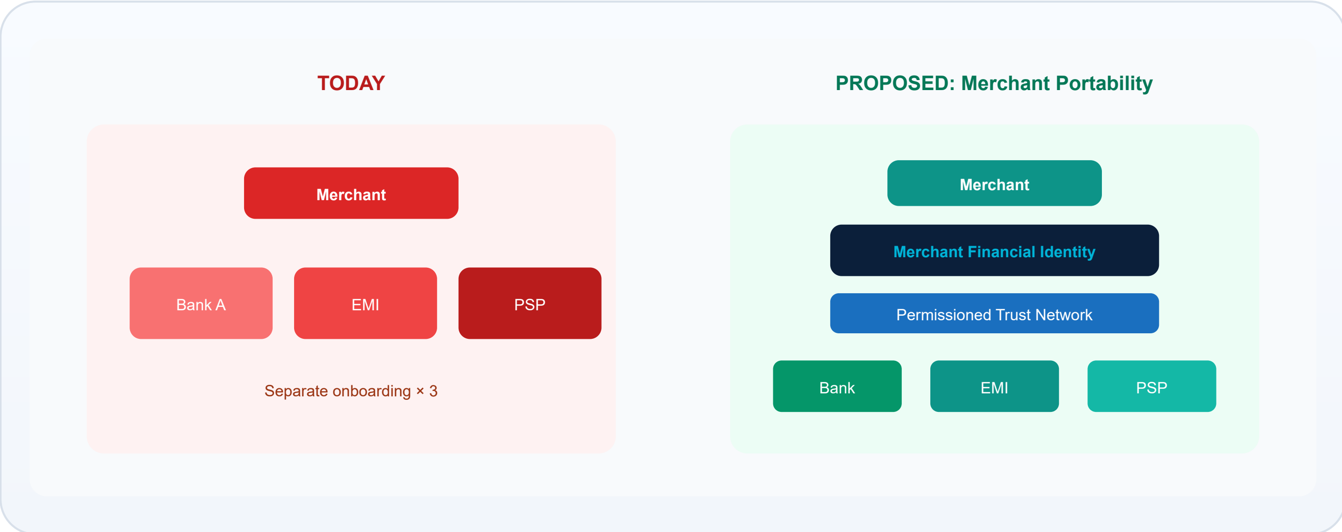


**Figure 13.** Merchant portability — one identity, many permissioned relationships.

# 12 Transaction Monitoring

PROPOSED ARCHITECTURE

The same trust layer could support AML/CFT workflows by linking customer identity, merchant identity, risk profiles and auditable investigation records.

**Important:** this paper does *not* claim blockchain automatically detects money laundering. Detection comes from rules, analytics and human investigation. The ledger can strengthen provenance and integrity of evidence trails — consistent with broader international discussions of DLT as an integrity/provenance tool rather than a magic compliance engine.[8][9]

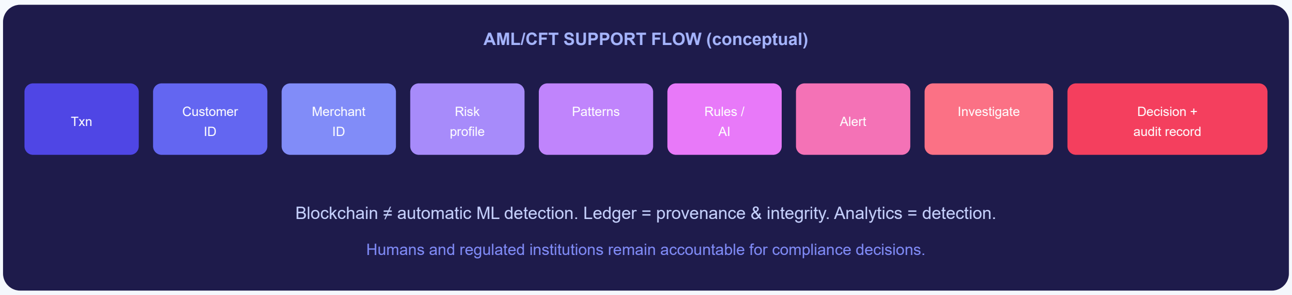


**Figure 14.** AML/CFT support flow — analytics detect; ledger records provenance.

# 13 Decentralized Banking vs Decentralized Trust

Pakistan should likely pursue a **regulated federated model**, rather than permissionless DeFi, for core banking trust.

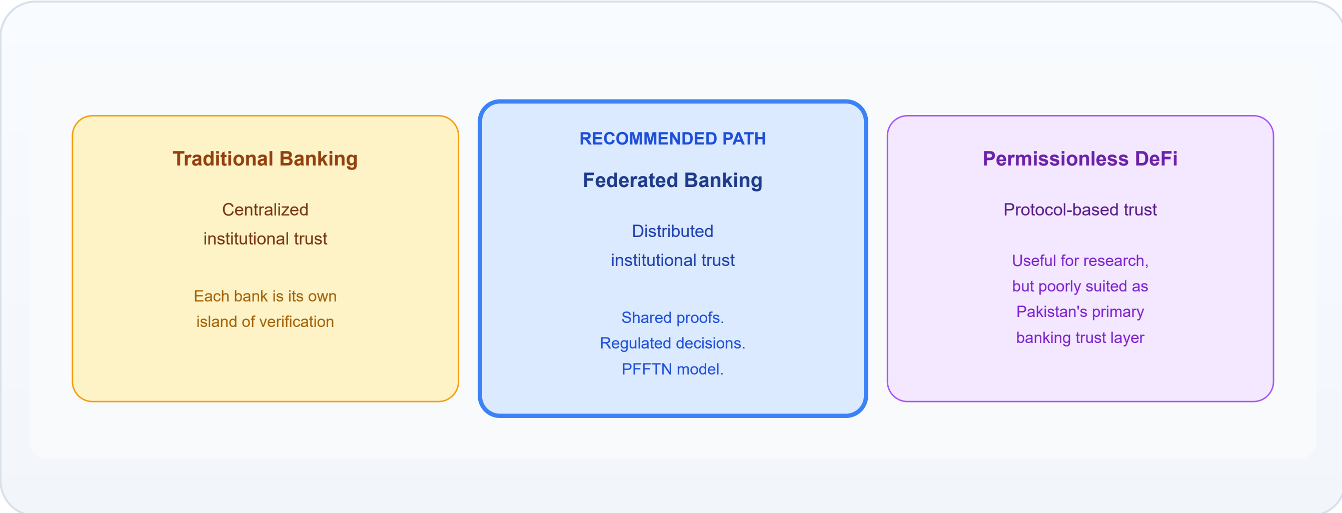


**Figure 15.** Three trust models — federated institutional trust is the practical middle path.

| Dimension | Traditional | Federated (PFFTN-style) | Permissionless DeFi |
|---|---|---|---|
| Trust source | Institution | Institutions + shared proofs | Protocol / code |
| Who can join | Licensed only | Permissioned participants | Anyone with a wallet |
| KYC/AML fit | **Native** | **Designed for** | **Often weak / external** |
| Customer redress | **Clear** | **Designable** | **Often unclear** |

| Dimension | Traditional | Federated (PFFTN-style) | Permissionless DeFi |
| --- | --- | --- | --- |
| Pakistan fit (core banking) | **Works, costly duplication** | **Strong conceptual fit** | **Poor primary fit** |

# 14 Governance

PROPOSED ARCHITECTURE

Potential participants could include SBP, banks, EMIs, PSOs, PSPs, MFBs, authorized identity providers and government verification entities.

Governance must cover: node participation, validator eligibility, data access, consent, revocation, dispute resolution, liability allocation, cybersecurity baselines, independent audit and regulatory supervision.

### SBP need not operate every node

Alternatives could include: industry consortium operators under SBP rules; hybrid models with supervisory nodes; or specialized utilities. This paper does *not* claim SBP should run all infrastructure.

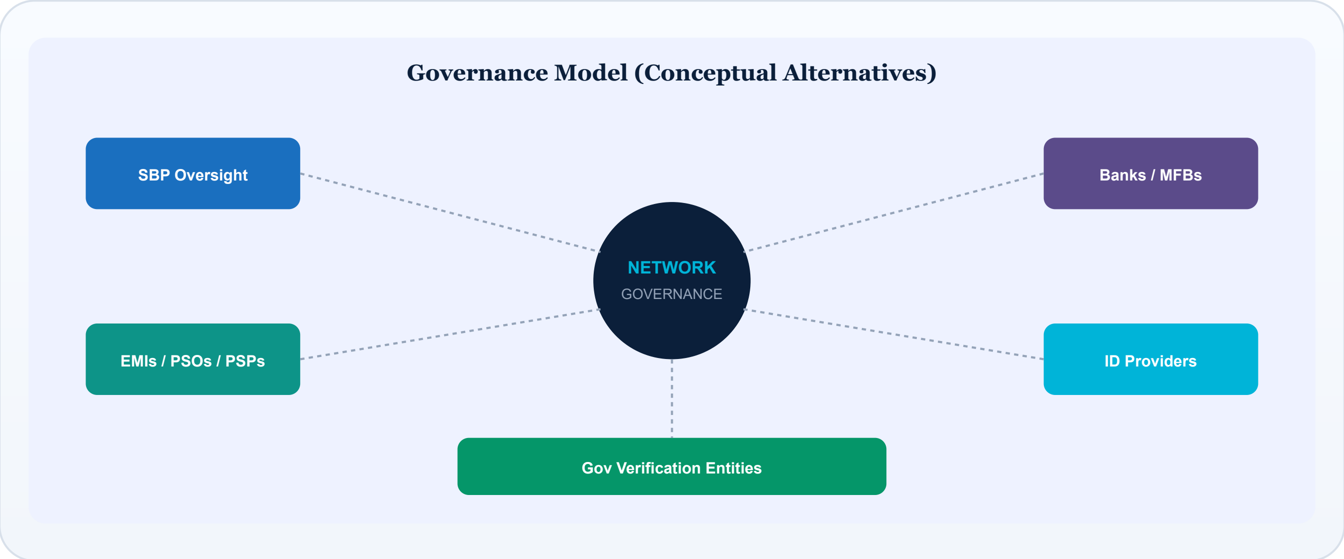


**Figure 16.** Conceptual governance constellation — oversight without assuming a single operator of all nodes.

# 15 Privacy

PROPOSED ARCHITECTURE

**Blockchain does not mean putting everyone's data on a public ledger.**

Privacy tools for a PFFTN-style design could include: permissioned DLT, zero-knowledge proofs where appropriate, selective disclosure, hashing, encryption, off-chain storage, credential revocation, consent management, role-based access and data minimization. International FMI/DLT research similarly emphasises restricting sensitive data visibility on shared ledgers.[8] [10]

### Design principle

If a teenager can screenshot it from a blockchain explorer, it should never have been personal data in the first place.

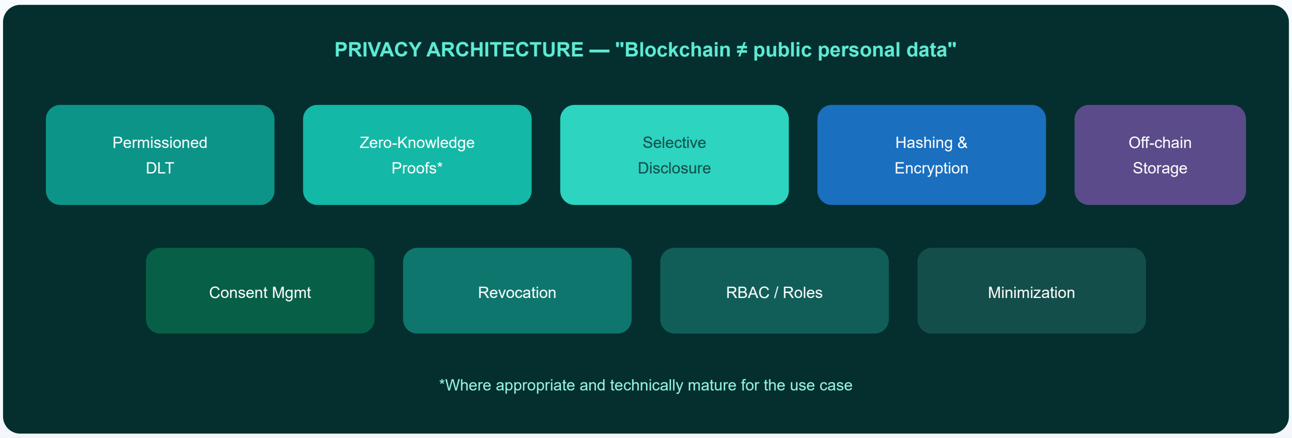


**Figure 17.** Privacy architecture toolkit for a permissioned financial trust network.

# 16 Security

PROPOSED ARCHITECTURE

Threats include private-key compromise, digital-signature abuse, identity theft, credential theft, node compromise, insider threats, smart-contract vulnerabilities, validator collusion, data poisoning, false credentials and failed recovery mechanisms. In permissioned networks, classic “51% attacks” morph into governance and collusion risks among validators.

**Decentralization does not automatically equal security.** Security is an operating discipline: key management, secure development, monitoring, audits, incident response and legal accountability.

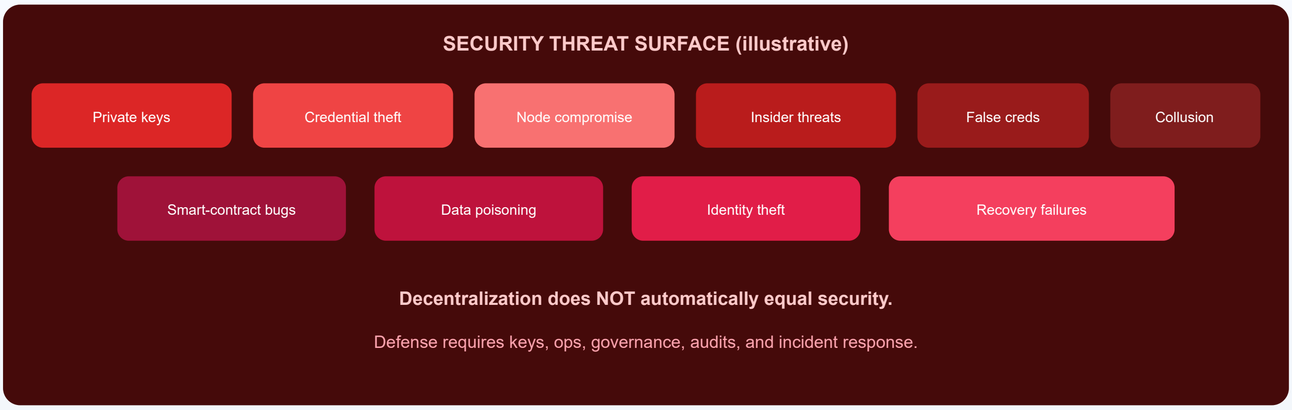


**Figure 18.** Illustrative security threat surface — decentralization is not a free security upgrade.

# 17 Economic Benefits

PROPOSED ARCHITECTURE

Potential benefits (qualitative; not guaranteed):

### Lower duplicated KYC cost

Reuse attestations instead of rebuilding dossiers.

### Faster account opening

Less paperwork friction for eligible customers.

### Faster merchant onboarding

Especially valuable for micro and SME merchants.

### Better inclusion UX

Fewer abandoned applications.

### Stronger auditability

Provenance of who attested what.

### Interoperability

Shared standards across participants.

### No invented savings figures

This paper does not invent precise rupee savings. Any future business case should use institution-specific cost studies. Hypothetical scenarios must be labelled as such.

# 18 A Teenager's Example

ILLUSTRATIVE STORY

Meet **Ali**, who is 18 and wants his first bank account. This story is fictional and simplified — meant to make the architecture click.

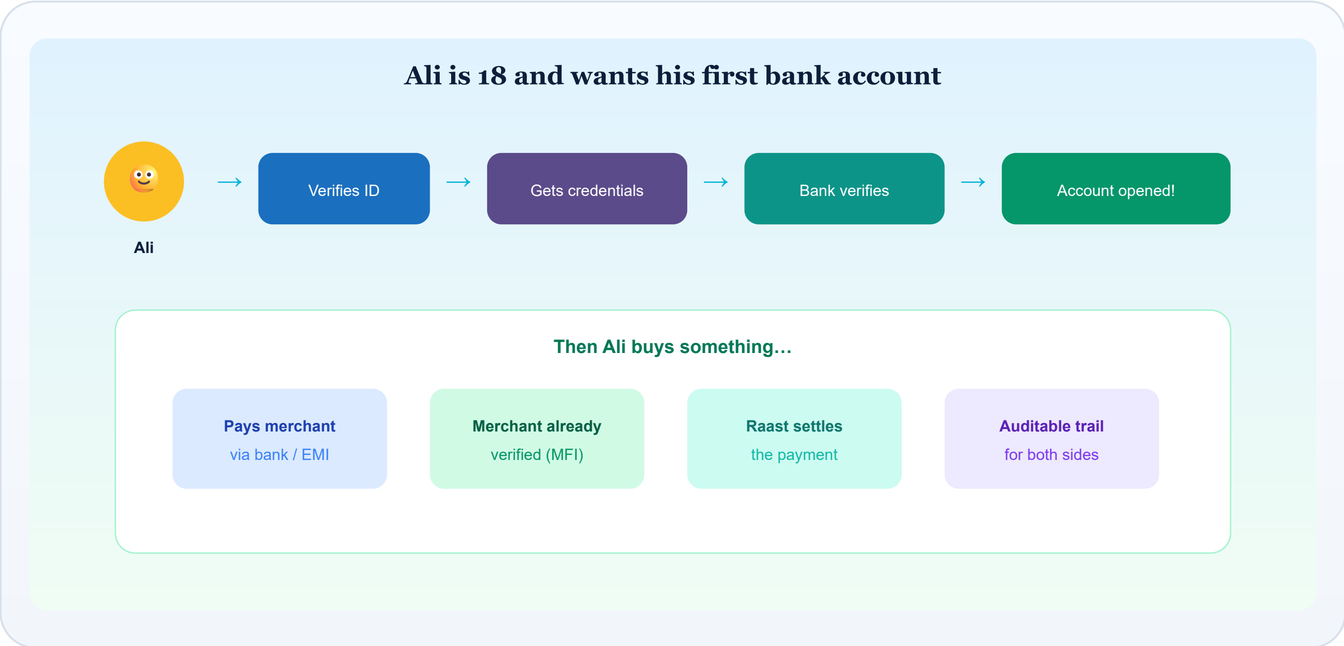


**Figure 19.** Ali's journey — identity credentials, bank decision, merchant payment, Raast settlement.

> **What Ali learns about blockchain:** it isn't a magic money app. In this design, it's more like a shared notary stamp for proofs — while banks still decide, and Raast still moves the payment.

# 19 The Future

FUTURE POSSIBILITY

If a federated trust layer matured, later extensions *could* include digital lending, SME financing, trade finance, remittances, insurance, supply-chain finance, government payments, tax verification, digital invoices, merchant credit scoring, open finance and embedded finance.

These are **not** current PFFTN capabilities — they are directional possibilities that would each need separate policy, legal and technical work.

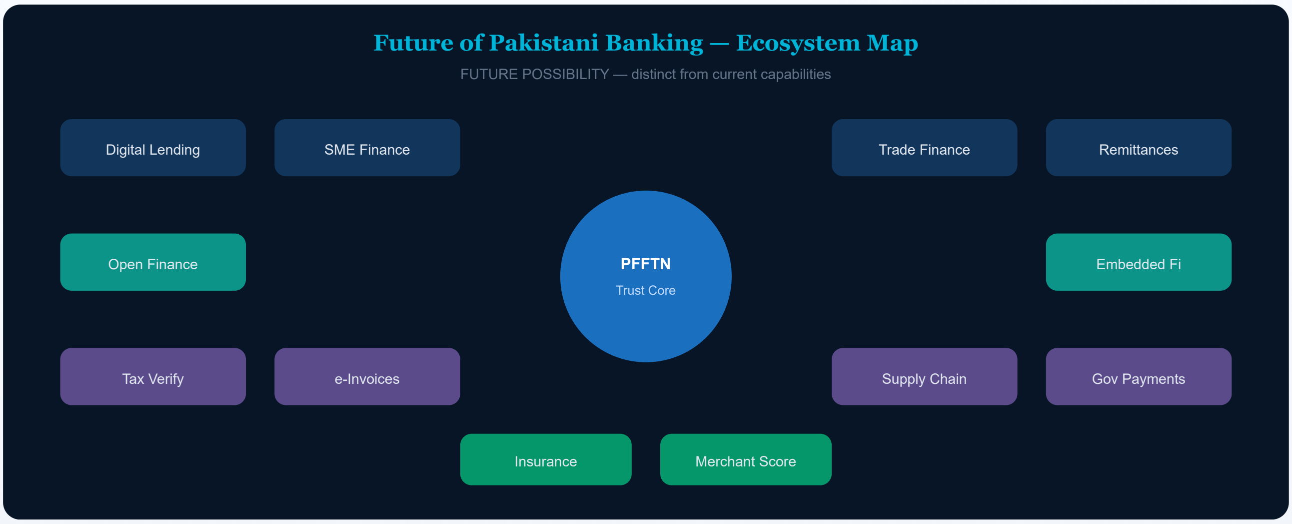


**Figure 20.** Future ecosystem map — possibilities beyond the core trust layer.

# 20 Implementation Roadmap

CONCEPTUAL ROADMAP

This is a conceptual sequence and would require SBP/regulatory approval and industry consensus. It is not a committed national programme.

PHASE 1

### Shared identity & credential standards

Define schemas, consent models, revocation, issuer rules, and alignment with Shared e-KYC concepts.

PHASE 2

### Merchant Financial Identity pilot

Limited participants; measure onboarding time, consent UX, dispute handling.

PHASE 3

### Integration with selected banks / EMI / PSP participants

Interoperability tests; liability playbooks; security assurance.

PHASE 4

### Raast P2M integration

Link merchant trust credentials to payment acceptance readiness — without replacing Raast.

PHASE 5

### National federated financial trust network

Broader participation under formal governance and supervision.

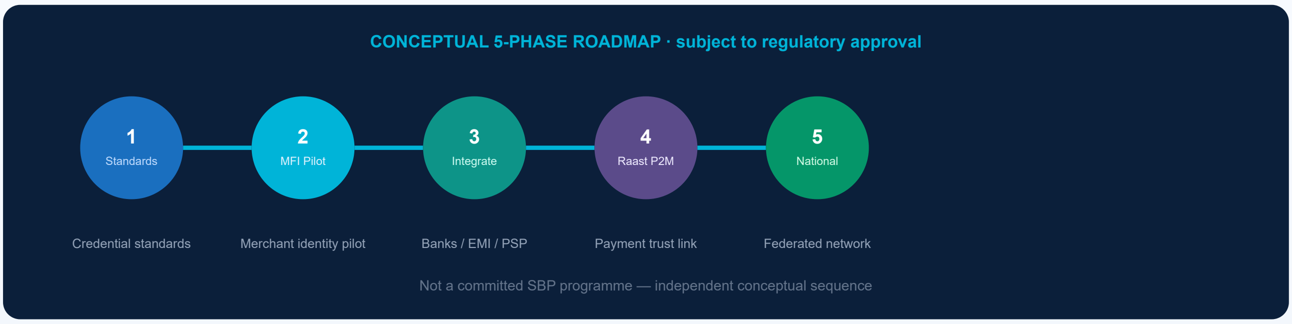


**Figure 21.** Five-phase conceptual roadmap — subject to regulatory approval.

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# Limitations and Open Questions

### Legal & liability

Who is liable if a credential is wrong? How does reliance work across institutions?

### Data protection

How do consent, retention and cross-institution access map to Pakistani law and evolving privacy norms?

### Inclusion

How do feature-phone users, biometrically excluded persons, and low-literacy customers participate fairly?

### Interoperability with Shared e-KYC

PFFTN must be designed as an extension path — not a competing silo.

### Operational resilience

What happens when nodes fail, credentials are revoked at scale, or a validator is compromised?

### Economics

Who pays for verification? How are free-riders prevented without recreating friction?

# Glossary

**AML/CFT**

Anti-Money Laundering / Combating the Financing of Terrorism.

**CDD**

Customer Due Diligence — understanding who a customer is and the nature of the relationship.

**DLT**

Distributed Ledger Technology — a shared, synchronised ledger across multiple parties.

**EDD**

Enhanced Due Diligence — deeper checks for higher-risk situations.

**EMI**

Electronic Money Institution.

**KYC**

Know Your Customer — identifying and verifying a customer.

**MFI (in this paper)**

Merchant Financial Identity — proposed reusable merchant trust profile (not "microfinance institution").

**PFFTN**

Pakistan Federated Financial Trust Network — the conceptual architecture proposed here.

**PSO / PSP**

Payment System Operator / Payment Service Provider.

**Raast**

Pakistan's instant payment system operated in the SBP ecosystem.

**Raast P2M**

Person-to-Merchant payment acceptance via QR, Alias, IBAN and Request to Pay.

**Verifiable credential**

A digital proof that a fact was verified by an issuer, presentable without always sharing raw documents.

# About the Author


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Ammar Jamshed is a data scientist and banking technology professional working at the intersection of data science, artificial intelligence, automation, financial technology and digital transformation.

He holds an MSc in Data Science from the University of London and has worked across banking, data science, AI, education and technology. His public profile identifies him as a data science and banking professional, with experience in data science, AI, automation, business analytics, consulting and education.

He has also been recognized through data-science and technology-related awards and programs, including recognition connected with Kaggle mentoring and the Pistoia Alliance Young Data Scientist of the Year 2025 shortlist.

He is the author of **“INCOHERENCES BETWEEN DATA AND ANALYSIS”** and founded Coursemon, an education technology initiative focused on technology and career learning.

LinkedIn profile

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**Again, for clarity:** this paper is independent research and conceptual work. It does not represent the views, policies, recommendations or endorsement of any employer, bank, regulator, government institution, client, partner or organization with which the author may be associated.

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Primary regulatory claims rely on SBP circular titles, dates and URLs. Where circular PDF body text could not be machine-extracted in this environment, secondary reputable reporting is cited for platform characteristics and labelled accordingly. No fabricated statistics are included.

**THE FUTURE IS NOT BANKS WITHOUT TRUST.**

IT IS BANKING BUILT ON SHARED, VERIFIABLE TRUST.

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